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Original Article

Digital twin for leak detection and fault diagnostics in gas pipelines: A systematic review, model development, and case study

Wahib A. Al-Ammari [a], Ahmad K. Sleiti [a,*], Mohammad Azizur Rahman [b], S. Rezaei-Gomari [c], I. Hassan [d], R. Hassan [e]

[a] Department of Mechanical & Industrial Engineering, College of Engineering, Qatar University, Doha, Qatar
[b] College of Science and Engineering, Hamad Bin Khalifa University, Education City, Doha, Qatar
[c] School of Computing, Engineering, and Digital Technologies, Teesside University, Middlesbrough, UK
[d] Petroleum Engineering Department, Texas A&M University at Qatar, Qatar
[e] Petroleum Engineering Department, Texas A&M University College State, USA

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ABSTRACT

Leak detection in oil and gas pipelines plays a crucial role in ensuring flow assurance. Numerous analytical and experimental methods are employed in the industry for leak detection. However, many of these techniques are costly or suffer from poor performance, including high false alarm rates. This research aims to develop a robust digital twin (DT) to address these issues. A comprehensive review is conducted, focusing on evaluating the performance of existing leak detection methods, including DT-based models. Furthermore, an overview of machine learning techniques used in the development of pipeline DT models is provided. The review reveals that current DT models are primarily focused on leak detection but do not adequately identify leak size and location. This study proposes a more comprehensive DT model capable of detecting pipeline abnormalities, such as leaks, equipment failure, and damage. The proposed DT is applied to a real-field gas pipeline case study to validate its feasibility. The results demonstrate that the developed DT model successfully detects leaks with a zero false alarm rate and accurately identifies leak size and location with an absolute relative error of less than 3.21 %. This work serves as a reference for future DT development and research in pipeline monitoring.

1. Introduction

Utilizing pipelines for natural gas logistics stands out as the most efficient, practical, and economical option. The majority of these pipelines are buried beneath the ground, traversing both commercial and residential areas. However, with the passage of time, corrosion, material defects, environmental factors, and abrasion pose inevitable threats to the pipelines' network integrity, leading to potential leakage failures. Leakage failures can result in severe environmental damage, harm to human well-being, and significant economic losses as outlined in Table 1. The severity of these consequences is particularly damaging when leaks go undetected for extended periods of time. Hence, the timely identification of leaks, or even predicting potential leaks in pipelines, is paramount to mitigating the impact of these hazards.

Pipelines represent closed systems encompassing physical components, computer systems, telecommunication systems, and the pipelines themselves. Diagnosing pipelines is not just about identifying system faults but also about continuously monitoring and detecting problems in a timely manner. Various methods are currently used to detect and locate leaks in pipelines. These methods can be categorized into three groups: exterior, visual/biological, and interior/computation-based methods [3,4]. Exterior methods involve the physical detection of escaping substances using external sensors [5]. Some of these methods include fiber optic sensors [6,7], acoustic sensors [8,9], and ground penetration radar [10]. These sensors are placed on the exterior of the pipeline and are used to detect changes in pressure, temperature, or sound waves that may indicate a leak. Fiber optic sensors use light to detect changes in temperature and pressure [11], while acoustic sensors detect sound waves generated by the escaping substance [12]. Ground penetration radar uses radio waves to detect changes in soil conductivity caused by the escaping substance, which can help locate the leak [13]. Visual/biological methods involve inspection by humans, trained dogs, and drones [14]. Humans and trained dogs can visually inspect pipelines on the ground for signs of leaks, while drones can be used to inspect pipelines from the air [15]. These methods are particularly useful in

* Corresponding author.
E-mail address: asleiti@qu.edu.qa (A.K. Sleiti).

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Nomenclature

Abbreviation

AE	Acoustic emission
AI	Artificial intelligence
AS	Anomaly score
ANN	Artificial neural network
ARE	Absolute relative error
BS	Backstepping
CFD	Computational fluid dynamics
CNN	Convolutional neural network
CVAE	Conditional variational auto-encoder
DDM	Data-driven model
DP-EEMD	Dual Pearson-ensemble empirical mode decomposition
DRSN-CW	Deep residual network with channel-wise
DT	Digital twin
ELMD	Ensemble local mean decomposition
EMD	Empirical mode decomposition
FAR	False alarm rate
FDR	Fault detection rate
FDT	Fuzzy digital twin
FE	Finite element
FGR	Fuzzy Gaussian regression
GP	Gas pipeline
GPLA	Gas pipeline leakage audio
GPR	Ground penetrating radar
GPS	Global positioning system
GR	Gaussian regression
HF	High frequency
HLP	Hazardous liquid pipelines
IoT	Internet of things
KPCA	Kernel principal component analysis
LD	Leak detection
LD&L	Leak detection and localization
LDM	Leak-detection model
LLE	Local linear embedding
LMD	Local mean decomposition
LSTM	Long short-term memory
ML	machine learning
MSE	Mean square error
NGP	natural gas pipelines
OP	Oil pipeline
OPELM	Optimally pruned extreme learning machine
OPEX	Operational expenses
PP	Physical pipeline
PPA	Pressure point analysis
PSO	Particle swarm optimization
ROM	Reduced order model
RSVM	Reduced support vector machine
SNR	Signal-to-noise ratio
SVM	Support vector machine
TL1DCNN	Transfer learning one-dimension convolutional neural network
TTBTs	Transient test-based techniques
VMD	Variable mode decomposition
VP	Virtual pipeline
WT	Wavelet transform

Table 1
Examples of human and financial losses due to leakage in oil and gas pipelines [1,2].

Year	Location	Major losses	Remark
2010	China	$10 million	Dalian Pipeline → rupture → fire → explosion → affected two crude oil pipelines
2014	India	22 fatalities and 37 injuries	Pipeline in Andhra Pradesh is blasted
2014	Malaysia	$1 million	Gas pipeline explosion at PETRONAS
1994–2013	USA	745 major accidents, 278 deaths, 1059 injuries, $110 million	The reported numbers herein are for gas distribution only.

detecting leaks in hard-to-reach areas or areas that are inaccessible to external sensors [16]. Interior/computation-based methods involve monitoring flow parameters using algorithms or models [17]. These methods include pressure point analysis, negative pressure waves, and dynamic modeling [18]. Pressure point analysis involves monitoring pressure at various points along the pipeline to identify changes that may indicate a leak [19]. Negative pressure waves involve generating a pressure wave in the pipeline and monitoring the time it takes for the wave to travel from one end of the pipeline to the other [20]. Any deviation from the expected travel time can indicate a leak. Dynamic modeling involves creating a computer model of the pipeline and using it to simulate flow conditions [21]. This method is particularly effective in detecting leaks with high accuracy and overcoming the limitations of data acquisition. Detailed comparisons between the features and limitations of the aforementioned leak detection methods are provided in some valuable review studies as in [22,23]. A qualitative analysis of the aforementioned leak detection techniques is presented in Table 2.

Model-based leak detection is a prominent approach in software-based techniques. It relies on two crucial components: the creation of a pipeline model based on process data or fundamental principles and the establishment of a threshold or boundary for normal operation. Creating accurate physical models for pipelines is challenging due to their complexity, lack of closed-form solutions, and the need for experienced personnel. As a result, data-driven models, which do not rely on interactive mathematical relationships, have been proposed to address real-world issues where traditional modeling is impractical. However, the practical application of these methods is hindered by the need for high-quality and sufficient data, representativeness, and data integrity, which are not always available in real-world systems. Unfortunately, it is typically easier to obtain a substantial amount of normal data as opposed to limited datasets of pipeline anomalies over the pipeline's lifetime. The disconnect between the physical pipeline and virtual data model results in fragmented, isolated, and stagnant data, making it difficult to achieve the accuracy required for real-time applications. Therefore, one of the key challenges in model-based methods is to create a highly accurate virtual model of the pipeline that genuinely reflects its dynamic behavior throughout its entire lifespan and can be updated in real-time.

As a novel approach, the digital twin (DT) emerges as a solution for addressing the challenges in cyber-physical integration [24]. Leveraging mathematical and data-driven models, it offers valuable insights into the occurrence and progression of anomalies. Consisting of virtual and physical systems with connection data that links the virtual, and physical components, DT empowers operators to monitor operations within a virtual platform, thereby enhancing pipeline system security [25]. Functioning as an actual condition service tool, DT allows operators to simulate a range of scenarios, including failures, equipment damage in operation, and component wear [26]. Thus, the gas transportation department can increase fault tolerance and safety in the pipeline by seamlessly integrating the real pipeline, virtual pipeline, data, and

Table 2
Performance evaluation of leak detection methods for oil/gas pipelines.

Method	Adaptive ability	Localization accuracy	Response time	Sensitivity	Evaluation ability	Continuous monitoring	False alarm rate	Maintenance requirement	Cost
Acoustic emission (AE) method	×	●	●	●	▼	✓	▲	●	●
Cable method	✓	▲	▲	▲	▼	✓	▼	▲	▲
Cross-correlation analysis	✓	●	▲	●	▼	✓	●	●	●
Digital twin-based method (Expected)	✓	▲	▲	▲	▲	✓	▼	▲	▲
Diode laser absorption	✓	●	-	●	▼	×	●	▲	▲
Distributed optical fiber sensor	✓	●	-	●	▼	✓	●	▲	▲
Dynamic pressure transmitter method	✓	●	●	▲	▼	✓	●	●	●
GPS time label method	✓	▲	▲	●	▼	✓	▲	●	●
Ground penetrating radar (GPR)	✓	▲	▲	●	▼	×	▼	●	●
Harmonic analysis	✓	●	●	●	●	✓	▼	▲	●
Lidar system	✓	●	-	●	▼	×	●	▲	▲
Millimeter wave radar system	✓	●	-	●	▼	×	●	▲	▲
Negative pressure wave method	✓	▲	▲	●	▼	✓	▲	●	●
Neural network analysis	✓	▼	▲	▲	▼	✓	●	▲	▲
Pressure point analysis (PPA)	×	▼	●	●	▼	✓	▲	●	●
SmartBall method	✓	▲	▲	▲	▼	×	▼	▲	▲
Sniffer method	×	▲	▲	▲	▲	×	●	▲	▲
Sonar system method	×	▲	▲	▲	▼	✓	▼	▲	▲
Sound wave method	✓	●	●	●	▼	✓	▲	●	▼
Spectral imaging method	✓	●	●	●	●	✓	▲	▲	▲
Method	**Adaptive ability**	**Localization accuracy**	**Response time**	**Sensitivity**	**Evaluation ability**	**Continuous monitoring**	**False alarm rate**	**Maintenance requirement**	**Cost**
State estimation method	✓	●	▼	●	●	✓	▲	▲	▲
System identification method	✓	●	▼	●	●	✓	▲	▲	▲
Thermal imaging method	✓	●	●	●	●	✓	▲	▲	▲
Transient test-based techniques (TTBTs)	✓	●	●	●	●	✓	▼	▲	●
Ultrasonic flowmeter method	✓	▲	▲	▲	▲	✓	●	▲	●
Ultrasonic-guided wave method	✓	●	▲	▲	▼	✓	●	●	●
Ultrasonic method	✓	●	▲	▲	▼	✓	●	●	▼
Vapor sampling method	✓	▲	▲	▲	▼	×	▲	▲	▲
Volume/mass balance method	×	▼	●	▼	▼	✓	▲	▼	▼
Legend	✓= Can, × = Cannot,								
	▲= Good, Fast, Strong,								
	▲/▲= High, ▼ = Poor, Slow, Weak, ▼= Low,								
	Black color = General, Red color = Disadvantage, Blue color = Advantage								

knowledge through the DT-based framework. In addition, the DT contributes to the reduction of maintenance expenses by providing a sensible, dependable, and economical maintenance plan.

While several review studies analyze the various leak detection methods in oil and gas pipelines [3,22,27,28], the fault diagnostic and DT models for abnormal detection in natural gas/oil pipelines have not been mentioned in these reviews. In addition, these reviews lack the analysis of the machine learning techniques that are used as software-based leak detection methods or as an internal part of the DT leak detection model. Furthermore, the literature lacks a clear and complete analysis of applying the DT models to detect abnormalities in gas pipelines. Therefore, the main contributions of this research study are:

- Conducting a comprehensive review of various machine learning techniques that are used in the pipelines' leak detection models.
- Analyzing the current status of fault diagnostics using DT techniques in versatile engineering systems.
- Comprehensively assessing the DT models that were proposed for leak detection in gas pipelines.
- Proposing a new framework for the DT model that can detect different abnormal conditions in gas pipelines.
- Applying the proposed DT model to detect and localize leaks in a real-field gas pipeline as a case study.

A review of the machine learning techniques that are used in the leak detection models is introduced in Section 2. Then, studies on fault diagnostics using DT techniques are analyzed in Section 3. The available DT models for leak detection with the proposed DT model in this review are presented in Section 4. Then, a detailed case study on the application of the proposed DT model for leak detection in real-field pipelines is introduced in Section 5. Finally, the conclusions and recommendations of this review study are summarized in Section 6.

2. Current status of using machine learning techniques in pipeline leak detection

Table 3 presents a comprehensive overview of leak detection studies in pipelines, spanning from 2012 to 2023, and employing various machine learning (ML) techniques. Notably, the studies cover a diverse range of pipeline types, including gas pipelines (GP), oil pipelines (OP), and natural gas pipelines (NGP), with objectives ranging from simple leak detection (LD) to the more intricate LD&L (leak detection and localization). The methods employed showcase the evolution of ML applications, incorporating traditional techniques such as support vector machines (SVM) and artificial neural networks (ANN) to more advanced approaches such as convolutional neural networks (CNN), Gaussian Mixture Models, and combinations like AE+SVM. Inputs for the models vary, including pressure, flow rate, temperature, and AE, reflecting a holistic consideration of factors influencing pipeline behavior. Despite this diversity, reported accuracy levels demonstrate the effectiveness of these ML techniques in achieving high precision, with some studies reporting accuracies exceeding 95 %. These studies also highlight the importance of data sources, with researchers utilizing experimental setups, historical data, and simulations from software like OLGA, GPLA-12, and CFD. Furthermore, the integration of physical models with ML approaches is evident in some cases, emphasizing a hybrid methodology for improved accuracy and interpretability.

However, challenges related to data availability, especially historical data or realistic experimental setups, remain noteworthy. The varying levels of reported accuracy underscore the need for a nuanced approach, carefully considering the specific characteristics of the pipeline, the nature of the data, and the chosen ML technique. As technology continues to advance, the findings from these studies offer valuable insights for the development and implementation of effective leak detection systems in diverse pipeline scenarios.

Table 3
Summary of leak detection studies based on machine learning techniques.

Year	Refs.	Scope	Objective	Method(s)	Inputs	Source of data	Accuracy (%)
2012	[29]	GP	LD	Rough set theory + SVM	P, Q	-	95
2013	[30]	GP	LD&L	AE + ANN	AE	Exp.	99
2014	[31]	GP	LD	LMD + SVM	P	Exp.	89
2015	[32]	GP	LD	ANN	-	Exp.	96
2017	[33]	GP	LD	ANN + IoT	Q	Exp.	-
2019	[34]	GP	LD	SVM + ANN	P, T, Q	Historical	-
2019	[35]	OP	LD	Gaussian Mixture Models	AE	Historical	-
2019	[36]	NGP	LD	CNN	AE	Exp.	70
2019	[37]	NGP	LD	AE+SVM	AE	Exp.	99
2019	[38]	NGP	LD	KPCA + CasSVM	-	Historical	97
2020	[39]	NGP	LD	ANN + Simple logic	P, Q	-	-
2020	[21]	NGP	LD	Hammerstein model + adaptive thresholds	P, Q	OLGA	95
2020	[40]	GP	LD	OPELM+BiLSTM	P	Exp.	99
2021	[41]	GP	LD	AE + k-NN	AE	Exp.	90
2021	[42]	OP/NGP	LD	PSO + SVM	NPW	Exp.	95
2021	[43]	NGP	LD	DRSN-CW	AE	GPLA−12	76
2021	[44]	GP	LD	TL1DCNN	P	Flowmaster	90
2021	[21]	NGP	LD&L	-	P, T, Q	OLGA	-
2021	[45]	NGP	LD&L	ANN	P, T, Q	OLGA	-
2022	[46]	WP	LD&L	ML + Physical models	P	Exp.	-
2022	[47]	NGP	LD	ANN, RSVM	-	Exp.	-
2022	[48]	NGP	LD	VMD + LLE + SVM	AE	Exp.	95
2022	[49]	GP	LD	ML		CFD	-
2022	[50]	OP/NGP	LD	FE + (ANN, Xgboost, Catboost)	-	-	-
2022	[51]	GP	LD	ANN	-	-	95
2022	[52]	-	LD&L	ML + Physical models	-	-	?
2022	[53]	-	LD&L	CNN	-	-	97
2023	[54]	GP	LD	EMD + DBN	P	Historical	98
2023	[55]	GP	LD	ML platform	AE	Exp.	99
2023	[56]	GP	LD	DP-EEMD + CNN	P, AE	Exp.	98

3. Current status of fault diagnostics using digital twin techniques

In this section, the general components of the DT model are explained (Section 3.1). Then, the current status of the DT models that were proposed for fault diagnostics purposes is highlighted in Section 3.2. After that, the available commercial DT models for fault diagnostics are discussed in Section 3.3.

3.1. Digital twin components

Fault diagnostics aim to pinpoint specific issues that are causing a system or device to malfunction. Therefore, in the case of fault diagnosis in oil and gas pipeline systems, leak size and location are predicted. DT architectures can be employed for fault detection in various industries [57], such as oil and gas, water management, and manufacturing. The concept of a DT involves creating a virtual replica or model of a physical system, which is continuously updated with real-time data to simulate and monitor the behavior of the actual system. When it comes to fault/leak detection in a pipeline, a DT architecture typically incorporates the following components (1) data Acquisition, (2) data integration, (3) modeling and simulation, (4) machine learning and AI, (5) visualization and analytics, (6) integration with control systems, and (7) continuous improvement. These DT model components are integrated and connected to drive the target applications. For instance, these components can be integrated based on the five-dimensional (5D) model (shown in Fig. 1) that is presented by Qi et al. [58]. They provided detailed information about the available technologies, and tools that enable the development of DT-based systems for various applications.

3.2. Digital twin models for fault diagnostics

Table 4 presents a summary of research papers showcasing the applications, methodologies, and contributions of DT technology in fault diagnostics of various engineering applications. Each paper focuses on a specific industry or system and aims to leverage DT technology for improved, performance, analysis, and prediction. Hu et al. [59] explore the petrochemical industry and propose a DT approach to enhance safety and controllability. Tang et al. [60] investigate precision spool valves, using DT technology to analyze assembly performance and uncover correlations with oil leakage. Wang et al. [61] focus on hydraulic systems, demonstrating the accuracy of their DT framework for fault diagnosis without the need for extensive historical fault data. Priyanka et al. [62] apply machine learning algorithms to develop a DT framework for predicting risk probability rates in oil pipelines. These studies emphasize the potential of DT technology to enhance safety, performance, and predictive maintenance in industrial settings. Xie et al. [63] concentrate on building HVAC systems, employing symbolic artificial intelligence techniques to identify informative sensory dimensions for building-specific faults. Dong et al. [64] introduce a DT-assisted feature fusion network to obtain reliable data for fault identification in hypersonic flight vehicles.

Wu et al. [65] focus on the water grid domain, utilizing DT technology to detect and localize anomaly events, such as pipe bursts or unauthorized water usage. These studies highlight the versatility of DT technology in different sectors, including building systems and infrastructure, and its potential to improve system performance, monitoring, and decision-making. Overall, these research papers collectively demonstrate the value of DT technology in various domains, providing insights into its potential applications, methodologies, and findings.

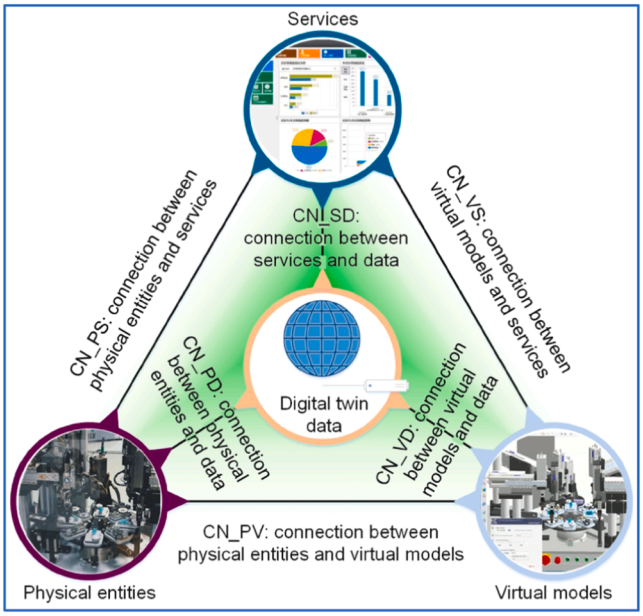


Fig. 1. Five-dimensional DT model proposed by Qi et al. [58].

Table 4
Proposed DT models for fault diagnostics in various engineering applications.

Refs.	Application	Target	Methodology	Contribution
[59]	Petrochemical Industry	Safety and Controllability	DT plug-in + decomposition + decoupling	Forecasting remaining service life, health status, and system reaction to incidents.
[60]	Spool Valve	Assembly Performance	Surface topography error analysis	Numerical relationship between oil leakage and mating parameters.
[62]	Oil Pipeline	Risk Probability Rate	Abnormal pressure analysis	Wireless networks for remote monitoring and risk rate prediction.
[61]	Hydraulic System	Predictive Maintenance	5D DT Framework	Achieving 89 % diagnosis accuracy without extensive historical fault data.
[63]	Building HVAC Systems	Informative Sensory Dimensions	Symbolic Artificial Intelligence	Filtering high-volume, diverse data in DT ecosystems.
[64]	Hypersonic Flight Vehicle	HFV Fault Data Generation	DT-assisted multiscale fusion network	Superior precision, recall, and measurement over other advanced methods.
[65]	Water Grid	Grid Operation and Management	Digital threads	Rapid anomaly detection/ localization, e.g., pipe bursts or unlawful usage.

4. Pipeline leak detection using digital twin and cyber-physical systems

This section presents a comprehensive review and comparison between the available DT-based gas pipeline leak detection techniques

(Section 4.1). Also, it highlights the features and limitations of the current commercial DT application (Section 4.2). Then, a new DT framework is proposed to detect pipeline abnormal conditions with a focus on leak detection and localization (Section 4.3).

4.1. Review of available pipeline digital twin models

In the open literature, a few DT-based methods for gas pipeline leak detection are introduced. In 2019, a conceptual design of a DT model for a subsea pipeline system was introduced by Bhowmik [66]. It consists of a machine learning (ML) algorithm-powered predictive maintenance model driven by field sensor data processing, and computational modeling. The computational part is initially generated using a finite element (FE) model, and it is calibrated using field sensor data that is obtained from the physical system (see Fig. 2(a)). After that, a ML model is used to forecast any variations in pipeline operation, including high pressure, slugging, or leaks, that may arise from abrupt changes in loading (see Fig. 2(b)). The author highlights the potential of this DT in minimizing maintenance expenses by strategically scheduling predictive future maintenance to prevent pipeline failures. However, it is noted that the study presents a broad concept lacking specific technical details about the DT components. Notably, no data is provided regarding the accuracy of the ML model.

For hazardous liquid pipelines (HLP), Zheng et al. [68] developed a DT model (depicted in Fig. 3) designed for leakage characterization. The DT predicts the leakage parameters using a conditional variational auto-encoder (CVAE), which leverages data from both downstream, and upstream meters. To assess the model's accuracy, stability, and general applicability, the authors validated it using four examples of pipeline leaks with varying lengths. The observed error between predicted and

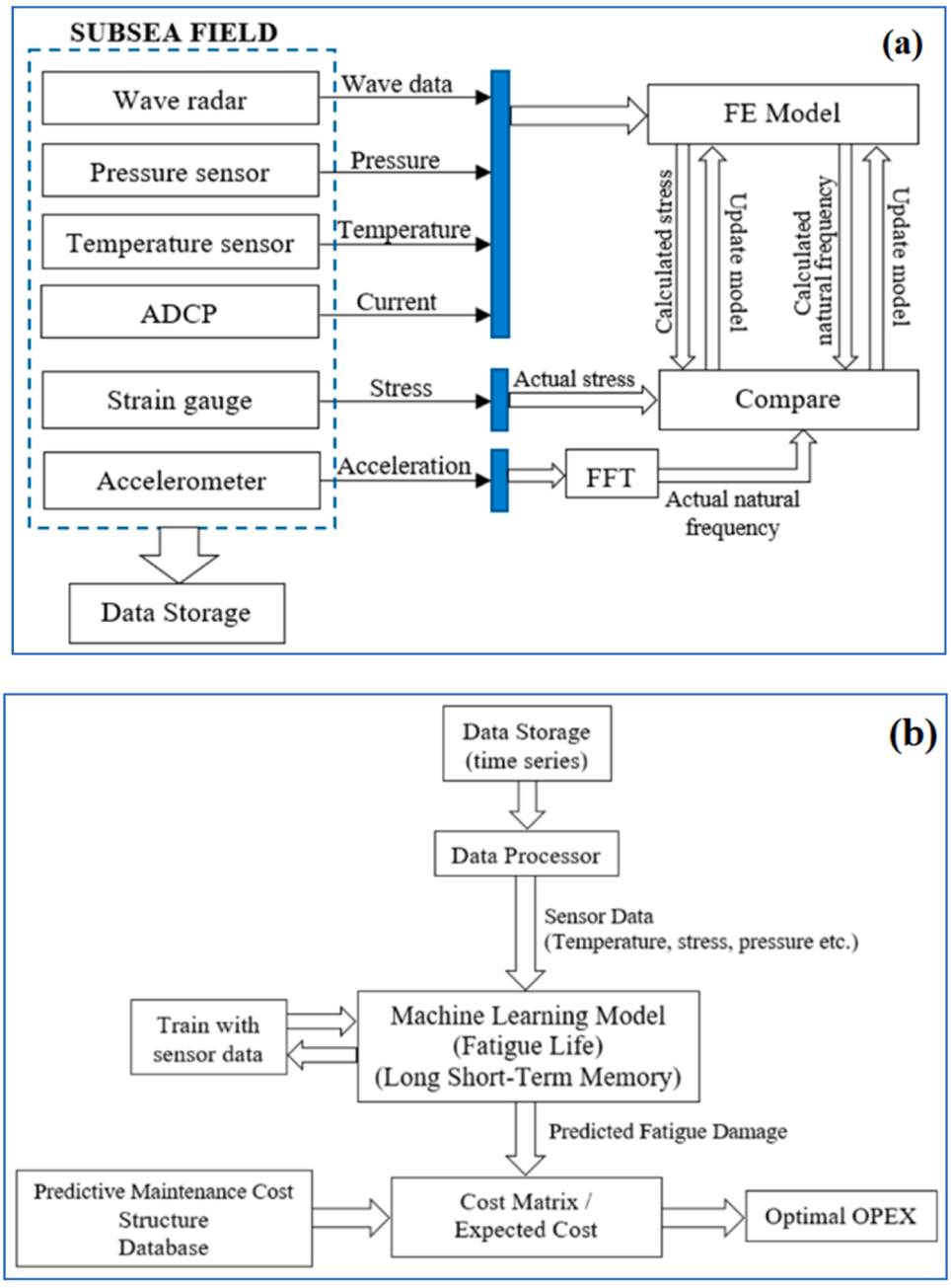


Fig. 2. Conceptual design of a DT for subsea pipeline proposed by Bhowmik [67]. (a) Calibrated FE model compared to field sensor data, and (b) data-driven predictive maintenance model. Figures (a) and (b) are adapted from [66].

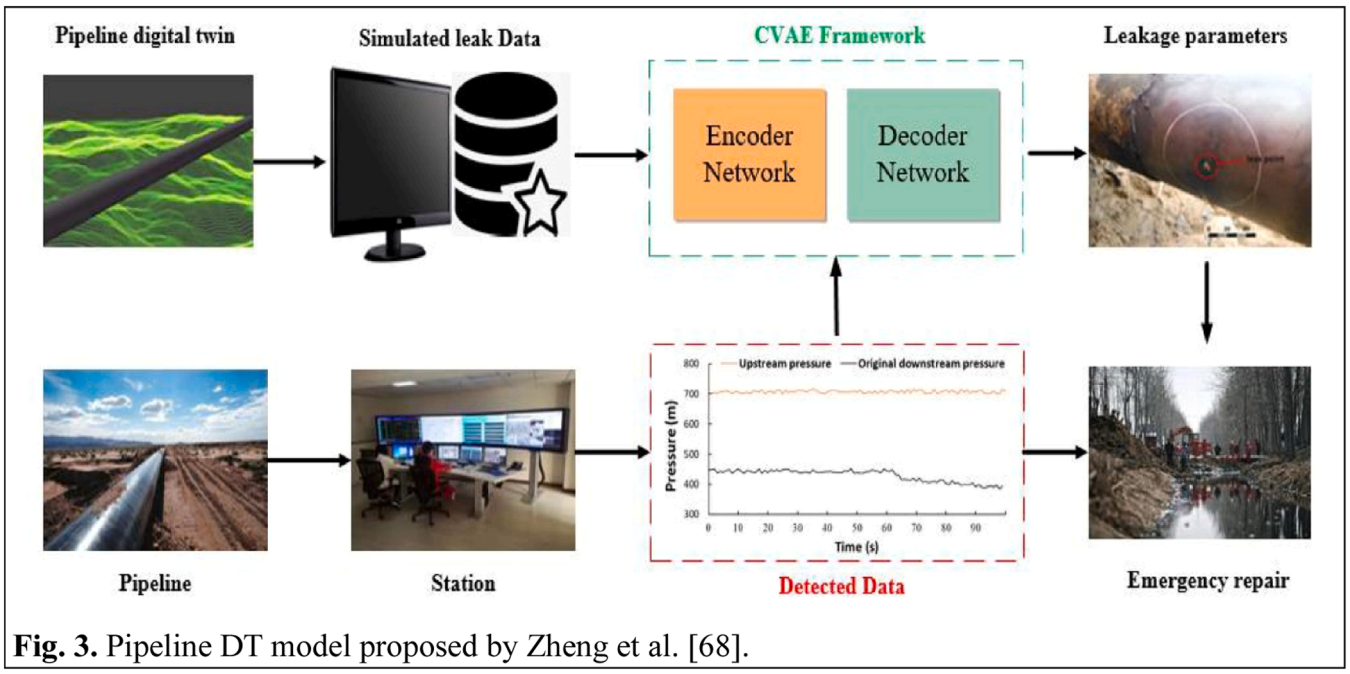


Fig. 3. Pipeline DT model proposed by Zheng et al. [68].

actual leakage parameters was found to be 3.9 %. However, the model is limited due to its dependency on the data of a particular pipeline. To facilitate the adaptation of this approach to other pipelines, they suggested the use of a transfer learning algorithm for transferring the model to new pipeline scenarios.

Piltan and Kim [69] presented a DT model for leak detection and localization, illustrated in Fig. 4(a). This model integrates a fuzzy DT, SVM, and backstepping (BS) observer. Initially, the fuzzy DT incorporates fuzzy Gaussian process regression (FGR) to model the flow rate under normal conditions. Subsequently, the DT is employed to estimate flow rate signals using FGR, BS, and fuzzy procedures, in both normal and abnormal pipeline conditions. The fuzzy DT also enhances localization accuracy as it plays a role in the data generation and calculation of residual data. An experimental setup (Fig. 4(b)) was built to evaluate the proposed approach's effectiveness The findings show a significant gain in defect localization accuracy, with the fuzzy DT outperforming conventional DT approaches by 25 %. A notable limitation of this method is its dependency on pressure/flow sensors, leading to reduced detection accuracy, especially in longer pipes. Thus, future studies could investigate the analysis of AE pipeline records using hybrid DT techniques, which presents a promising way to improve the effectiveness of leak identification.

Priyanka and Thangavel [70] introduced the concept of a DT in combination with classification architecture and feature extraction to precisely detect oil pipeline leakages (see Fig. 5). Their approach uses the K-L and ensemble local mean decomposition (ELMD) method for feature extraction. High-level product functions are extracted using ELMD, which estimates leak parameters by identifying influential clusters. Time-frequency features are processed using K-L divergence, and the resulting features are organized into a Markov chain to allow for precise classification using small datasets. The results of the experiments demonstrate how well ELMD interprets leakage signals and produces useful product functions. Data gathered from pressure transmitters with different leakage gaps is included in the study. Also, when combined with fast data transmission and the best possible downlink power, the DT can produce good results even at low signal-to-noise ratios (SNRs) with an accuracy of 91.23 %, surpassing the performance of SVM and NB methods.

Wang et al. [71] developed a DT model for pipeline leak detection using generated pressure signals. Online updating is used in this model to continuously update the data in the twin model that reflects the physical pipeline and its parameters. The Twinbuilder platform in ANSYS is then used to exhibit these updates in a real-time visualization model. In parallel, a pipeline system is constructed in the laboratory to validate the effectiveness of the developed model. As depicted in Fig. 6 (a), the pipeline system comprises a DN89 pipe (183 m), pressure sensors, and a controllable opening valve. The physical pipeline is simulated and visualized using a three-dimensional reduced order model (3D-ROM) as illustrated in Fig. 6(b). The pipeline data is classified and recognized using a Support Vector Machine (SVM), using 800 training datasets and 200 testing datasets. The outcomes of the experiments show an identification accuracy of more than 80 %, which paves the way for the development of more sophisticated DT pipeline leak detection algorithms.

Liang et al. [72] introduced a sophisticated DT for a pipeline using normal operating conditions to construct a data-driven virtual model (see Fig. 7). Using deep learning techniques, the DT pipeline independently recognizes underlying patterns in the physical pipeline's normal operating data. Additionally, the DT model has a data-driven leak detection method, which enhances the twin pipeline's capability to recognize abnormal working conditions. Subsequently, the DT pipeline model is deployed in a real-running natural gas pipeline, and analyses of effectiveness and performance are conducted. The results underscore the efficiency and performance of the proposed approach for status monitoring and anomaly detection in natural gas pipelines. The DT pipeline model is then implemented in an actual natural gas pipeline that is currently in operation, and performance assessments are carried out. The outcomes highlight the effectiveness of the suggested method for natural gas pipeline status monitoring and anomaly identification. However, it's noteworthy that this model is specifically designed for leak detection without providing leak localization. Moreover, its training is limited to a single leak condition, thereby restricting its ability to detect multiple leak scenarios.

As depicted in Table 5, the DT models designed for leak detection exhibit the following observations: (1) their primary emphasis lies in identifying the occurrence of a leak, with no provision for pinpointing its specific size and location; (2) these models display constrained experimental validation, resulting in diminished accuracy, particularly for

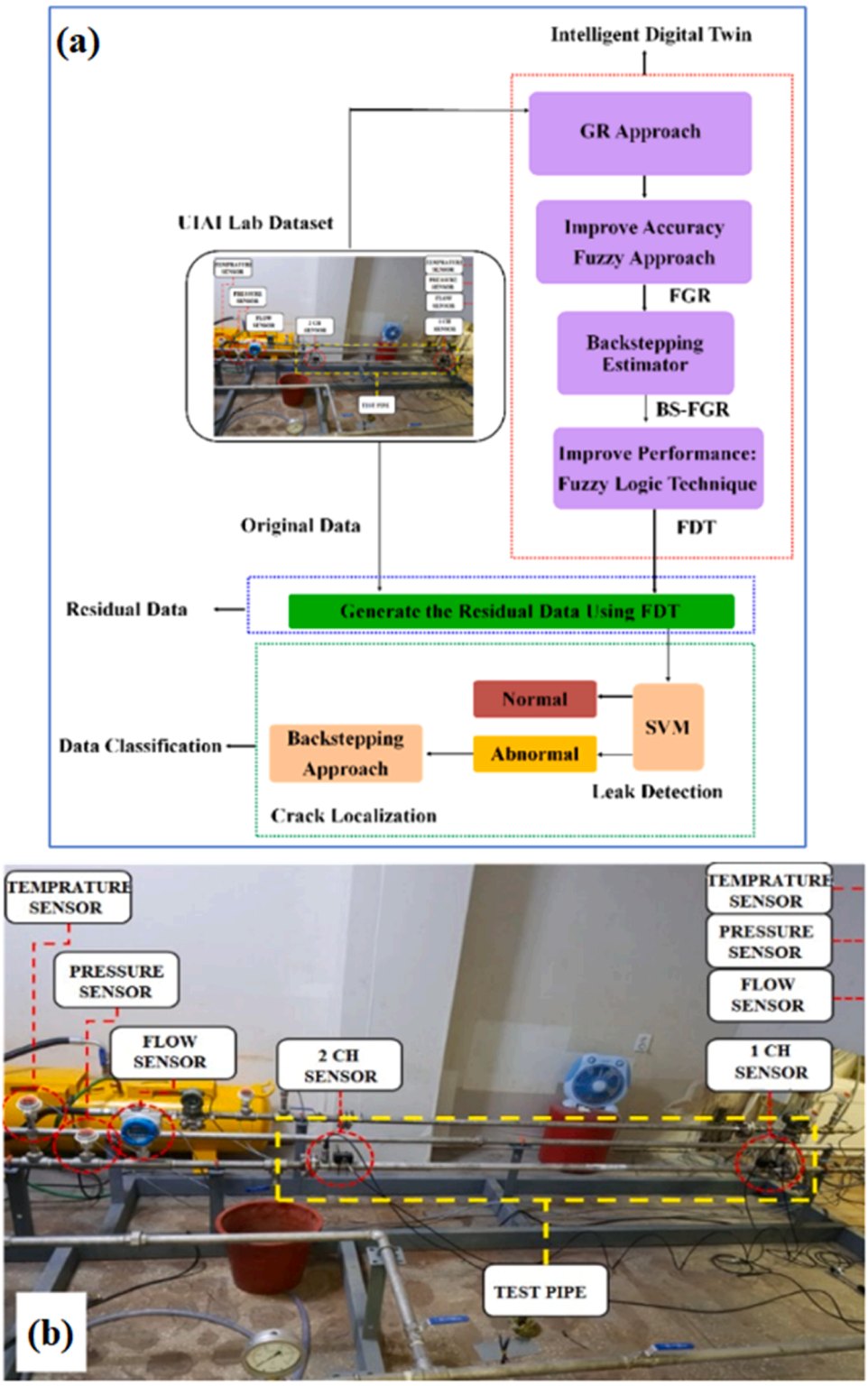


Fig. 4. (a) DT-based leak detection and localization model proposed by Piltan and Kim [69], and (b) experimental setup of UIAI lab that is used to test and validate the developed DT model.

extended pipelines; and (3) there is an absence of information concerning the experimental methodology employed to generate data related to leaks. Additionally, these models exhibit a deficiency in their capacity to detect various leak scenarios. Moreover, there is a notable absence of a DT specifically tailored for both the detection and localization of leaks in lengthy natural gas pipelines.

4.2. *Commercial pipeline digital twin tools*

A few commercial DT tools are currently used for leak detection as reported in Table 6. Although the available information about these tools is limited, these tools are used for leak detection, and further improvement is still needed to localize the leak and identify its size for different scenarios (such as single/multiple leaks, single/multi-phase

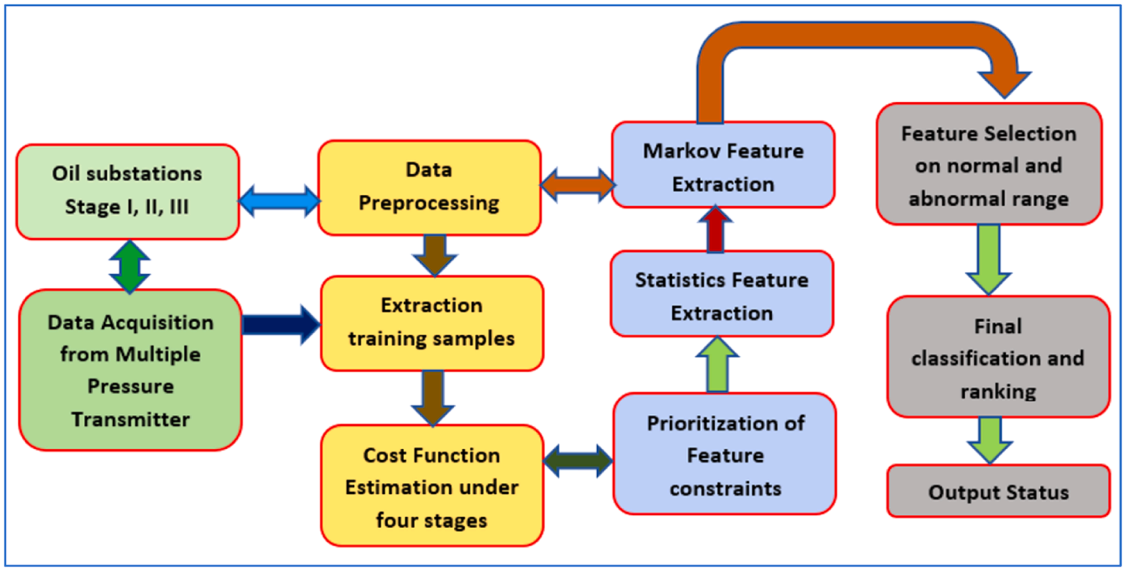


Fig. 5. Schematic presentation of the DT model proposed by Priyanka and Thangavel [70].

flow, onshore/offshore pipelines, oil/gas pipelines, transportation/distribution pipelines, etc.). Furthermore, the capabilities of these tools need to be extended to detect any abnormal conditions during the operation of the pipeline.

4.3. *Proposing pipeline digital twin model for multiple-leak detection and localization*

This section describes the framework of the proposed DT model in this review study. Unlike other DT models available in the literature, the proposed DT model is designed to detect different types of abnormal operating conditions including single/multiple leak conditions, and malfunction of the pipeline components (such as valves, and compressors). More importantly, it is designed to identify the leak size and location with a suitable action plane to be applied to the physical pipeline. The proposed DT model is developed based on the high-fidelity presentation of the physical pipeline in the virtual space using seven interactive parts which are: (1) data services, (2) data-driven models, (3) anomaly score (AS) calculation, (4) data analysis engine, (5) constrained condition library, (6) malfunction diagnostic techniques, and (7) historical DT library. Below is a detailed description of the DT framework and its interactive parts.

4.3.1. *Framework of the digital twin model*

The framework of the proposed pipeline DT is shown in Fig. 8. The pipeline DT model integrates the physical pipeline (PP) with the virtual pipeline (VP) as follows:

First, the measured data from the PP (mass flow, pressure, and temperature) are continuously collected by the data services platform (Section 4.3.2). From this platform, normal data sets are directed to the digital pipeline platform to train the VP model, which is referred to as the data-driven model (DDM, Section 4.3.3). The DDM will be trained to predict the flow parameters at the pipeline outlet based on the measured data at its inlet. Therefore, after training the DDM, the real-time inlet data ($\{I\}^{t}_{j,PP}$) will be directed to the VP to predict the virtual outputs ($\{O\}^{t+k}_{j,VP}$) (under normal operating conditions). These virtual outputs are compared with the real-time outputs ($\{O\}^{t+k}_{j,PP}$) by the residual between them, referred to as AS (Section 4.3.4). The AS represents the difference between the virtual outputs (that should be obtained if the pipeline works at normal conditions without any abnormalities) and the real-time outputs. If the virtual and real-time outputs are matched, this implies that the pipeline works normally without abnormalities. If there are significant differences between them, then this may indicate abnormalities that need further data analysis to be detected. To facilitate this process and to avoid false alarms, the AS is compared with the threshold value (Ψ) that was previously identified by statistical analysis of the pipeline normal data. If the AS is higher than Ψ, then the data are labeled as abnormal conditions. These abnormal data are directed to the data analysis engine (Section 4.3.5). In this engine, pre-trained models for abnormality detection will be used to identify the abnormal conditions that match the new feed data. Once the characteristics of the abnormal conditions are detected, they will be visualized in the digital model platform with an alarm and action plan to be implemented in the physical pipeline.

The proposed DT model fundamentally differs from existing DT-based methods in several key areas. First, it integrates a novel AS calculation technique that combines both statistical and machine learning-based thresholds, allowing precise detection of abnormal operating conditions. Second, the data-driven model incorporates a hybrid approach that leverages both physical simulation tools (e.g., OLGA and Pipesim) and machine learning algorithms, ensuring high adaptability to varying pipeline conditions. Third, the system architecture is designed to support scalability and generalizability by including a historical DT library and transfer learning techniques, enabling adaptation to pipelines of different lengths, diameters, and operational conditions. These features collectively enhance the model's capability to detect leaks with zero false alarms and localize leak size and location with minimal error.

While the proposed DT model primarily focuses on leak detection and localization, its modular architecture enables easy adaptation to other fault diagnostics, such as corrosion monitoring or flow assurance challenges. For instance:

- *Corrosion Monitoring:* The AS calculation module can be modified to include parameters like changes in wall thickness, pitting rates, or pipeline material degradation detected through ultrasonic sensors or smart pigging data. Machine learning models can then predict the extent and progression of corrosion.
- *Flow Assurance:* The data-driven model can be trained on flow regime data, wax deposition rates, and hydrate formation tendencies to identify and mitigate flow assurance issues. Simulation tools like OLGA or Aspen HYSYS can be integrated to model transient flow scenarios. These adaptations require minimal changes to the existing

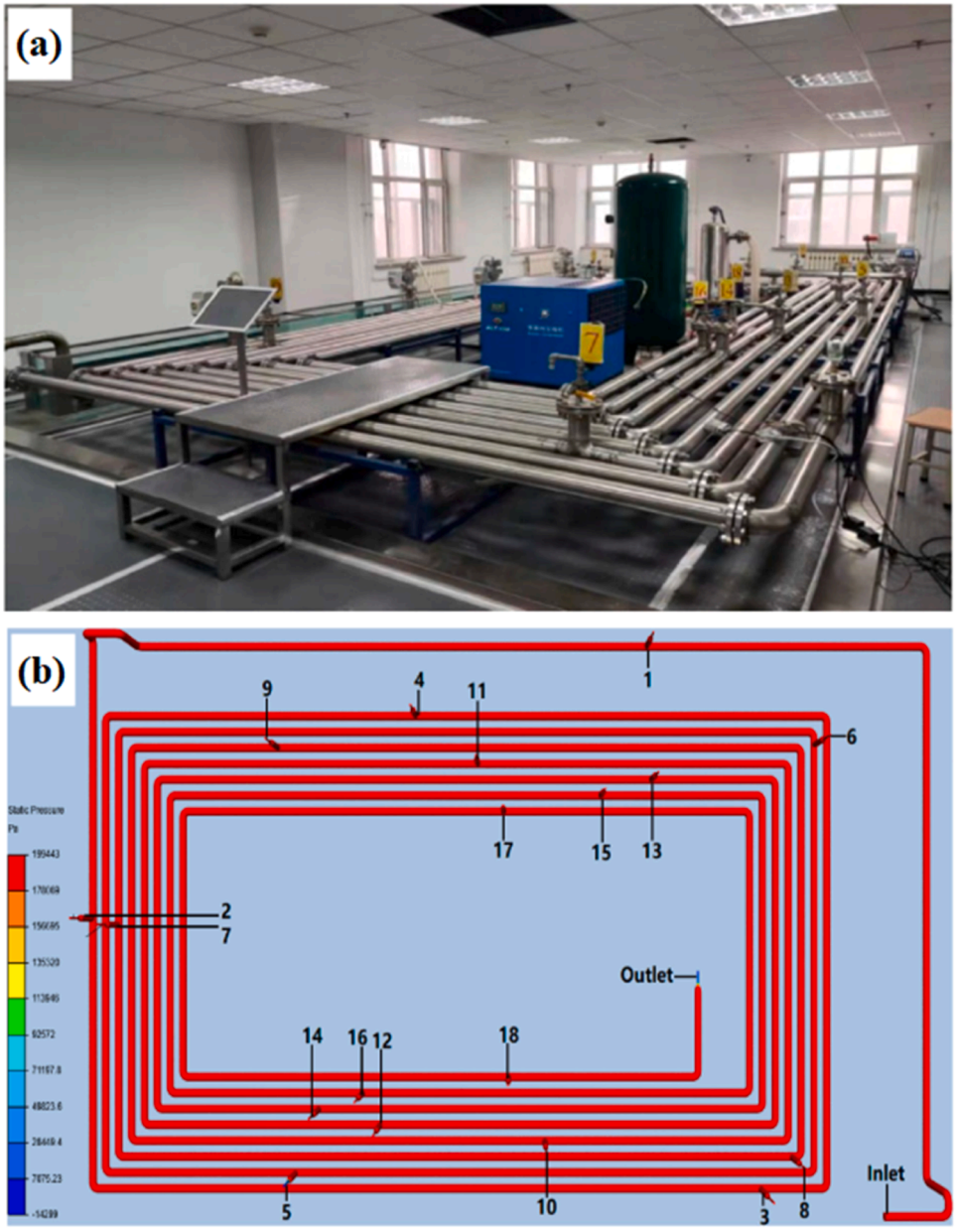


Fig. 6. (a) Physical pipeline system, and its (b) 3D ROM simulation and visualization for the DT pipeline leak detection model established by Wang et al. [71].

framework, leveraging its core capabilities to diagnose diverse pipeline abnormalities effectively.

4.3.2. Data services

DT must manage an extensive dataset encompassing physical, virtual, connection, service, and fusion data. Consequently, there is a need for thorough data preparation and collection. This process is vital for uncovering and comprehending the patterns of data evolution related to the natural gas pipeline's operation. It involves gaining insights into industry knowledge, interpreting changes in index data, understanding various operational conditions, and more. Data collection goes beyond retrieving operational data solely from the SCADA system and other sensing devices; it also involves acquiring data mappings based on previous business knowledge.

The absence of detailed descriptions regarding pipeline behavior during anomalies makes marking abnormal samples challenging. As a result, an imbalance exists between normal operation samples and abnormal ones. Typically, two approaches are employed to address this sample imbalance:

(i) Expanding the dataset of leakage samples through simulation experiments, such as building a pipeline simulation model using multi-phase flow simulation software or installing a leak experiment device on the physical pipeline to obtain leakage samples.
(ii) Balancing samples from both a data and algorithmic perspective may involve resampling datasets or applying new algorithms to enhance learning from minority class instances.

The proposed DT framework is designed for seamless integration with existing pipeline monitoring systems, leveraging real-time data from SCADA systems, IoT sensors, and other operational infrastructure. By utilizing modular architecture, the DT can ingest data from pressure, flow, and temperature sensors already deployed in pipelines. This data is processed through machine learning algorithms trained on historical and synthetic datasets to predict leak size, location, and operational anomalies.

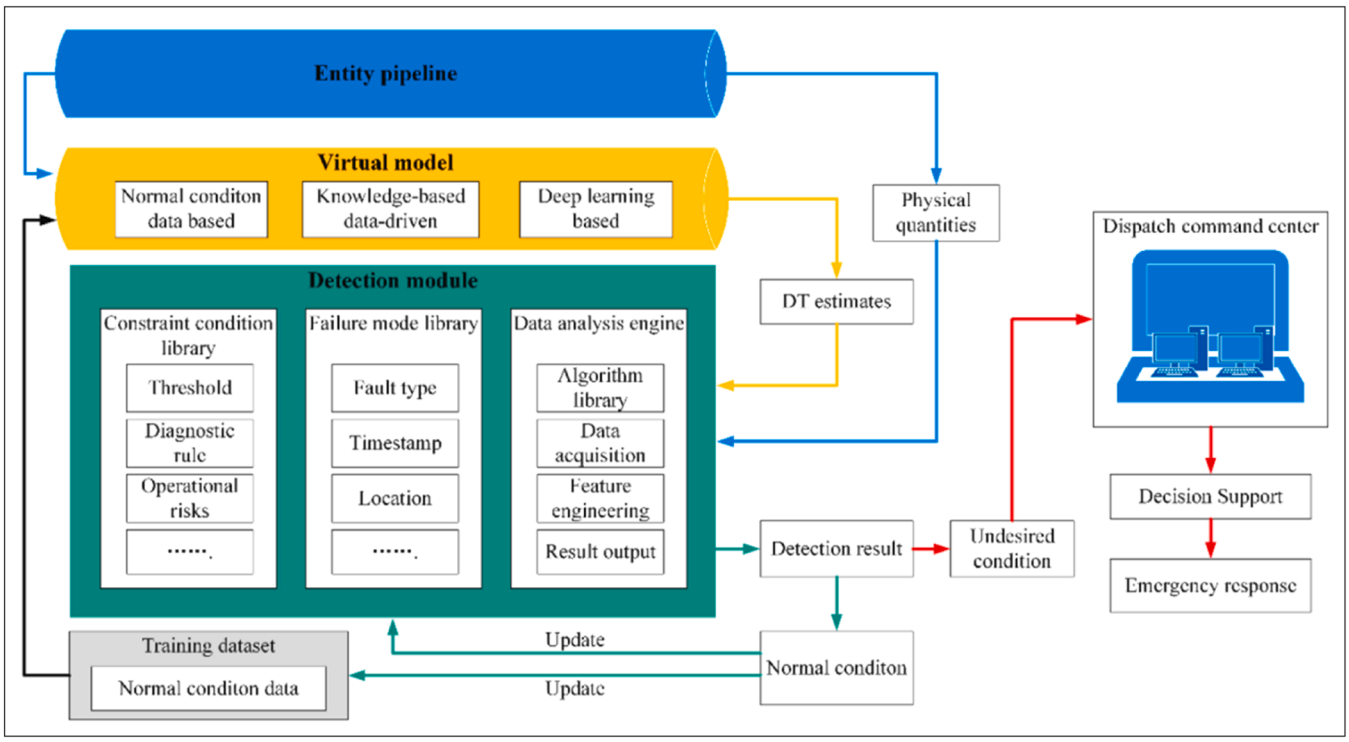


Fig. 7. Structure of the DT pipeline (proposed by Linag et al. [72].

Table 5
Features and limitations of the proposed DT models for pipeline leak detection.

Refs.	Virtual approach	Advantages	Shortcomings
[67]	FE + RNN	Reduce maintenance costs.	LD only.
[68]	CVAC	Error of 3.90 %.	Limited for liquid pipelines.
[69]	FDT + SVM + BS	25 % better than with conventional DT.	Inaccurate for long pipelines
[70]	ELMD + K-L	Identification accuracy of 91.23 %.	LD only.
[71]	3D ROM	Visualization model.	LD only.
[72]	Data-driven model	Validated experimentally.	LD only.

Table 6
Available commercial DT tools for leak analysis and detection in oil/gas pipelines.

#	Company	Software/Tool	Leak relevancy
1	Schlumberger [73]	OLGA Online Production Management System	Key events notifications (leaks, hydrates, surges, etc.) (limited for production wells).
2	Emerson [74]	Pipeline Manager	Provides pipeline leak detection, and leak localization.
3	Ansys [75]	Ansys Twin Builder	Allows engineers to create simulation-based DTs with Hybrid Analytics.
4	PipePatrol [76]	KROHNE	Leak detection based on RTTM.

Integration involves connecting the DT framework to the existing SCADA network using standardized protocols for data exchange. Advanced calibration mechanisms ensure compatibility with pipeline-specific conditions, such as variations in diameter, material, or operating pressures. Furthermore, the DT framework's adaptability enables dynamic updating of prediction models as new data is collected. This approach ensures scalability, allowing the framework to support pipelines of varying configurations while maintaining high predictive accuracy. The integration process is outlined in the supplementary material (Section S5), which provides systematic steps for adapting the DT framework to different pipeline systems

4.3.3. *Data-driven models*

Developing a data-driven model (DDM) for the pipeline based on normal operation data forms the VP part of the DT pipeline model. Denoising pressure, flow, and temperature signals before using them to train a virtual model for gas pipelines is essential to ensure the accuracy and reliability of the model. Several typical methods for denoising these signals are available, each with its features and limitations. Here are some common denoising methods for these signals:

The choice of denoising method depends on the specific characteristics of the signals, available data, and the goals of the model. Often, a combination of methods or an iterative approach is used to achieve the best results. It's essential to carefully evaluate the denoised signals' impact on the virtual model's performance and adjust the denoising process accordingly. Additionally, it's crucial to consider the trade-offs between noise reduction and the potential loss of important signal features. Usually, empirical mode decomposition (EMD) and wavelet decomposition (WL) are employed to denoise the original signal (Fig. 9).

After preparing normal operating data (healthy data that are free of any abnormalities), DDM can be developed based on machine learning methods (Section 2). For example, autoencoder-convolutional neural network (AE-CNN) techniques can be used to develop the DDM as proposed by Liang et al. [72]. As shown in Fig. 10, the input data (from the data service platform) are divided into vectors of particular length as y_{input}. Then, the features of y_{input} are captured by convolution processing and recovered to the reconstruction sequence $y_{recreate}$ by deconvolution process. However, this approach may require substantial computational resources. Other machine learning techniques that can be used to train the DDM with their features and limitations are presented in Table 8.

4.3.4. *Anomaly score calculation*

To facilitate the comparison between the outputs of the DDM from

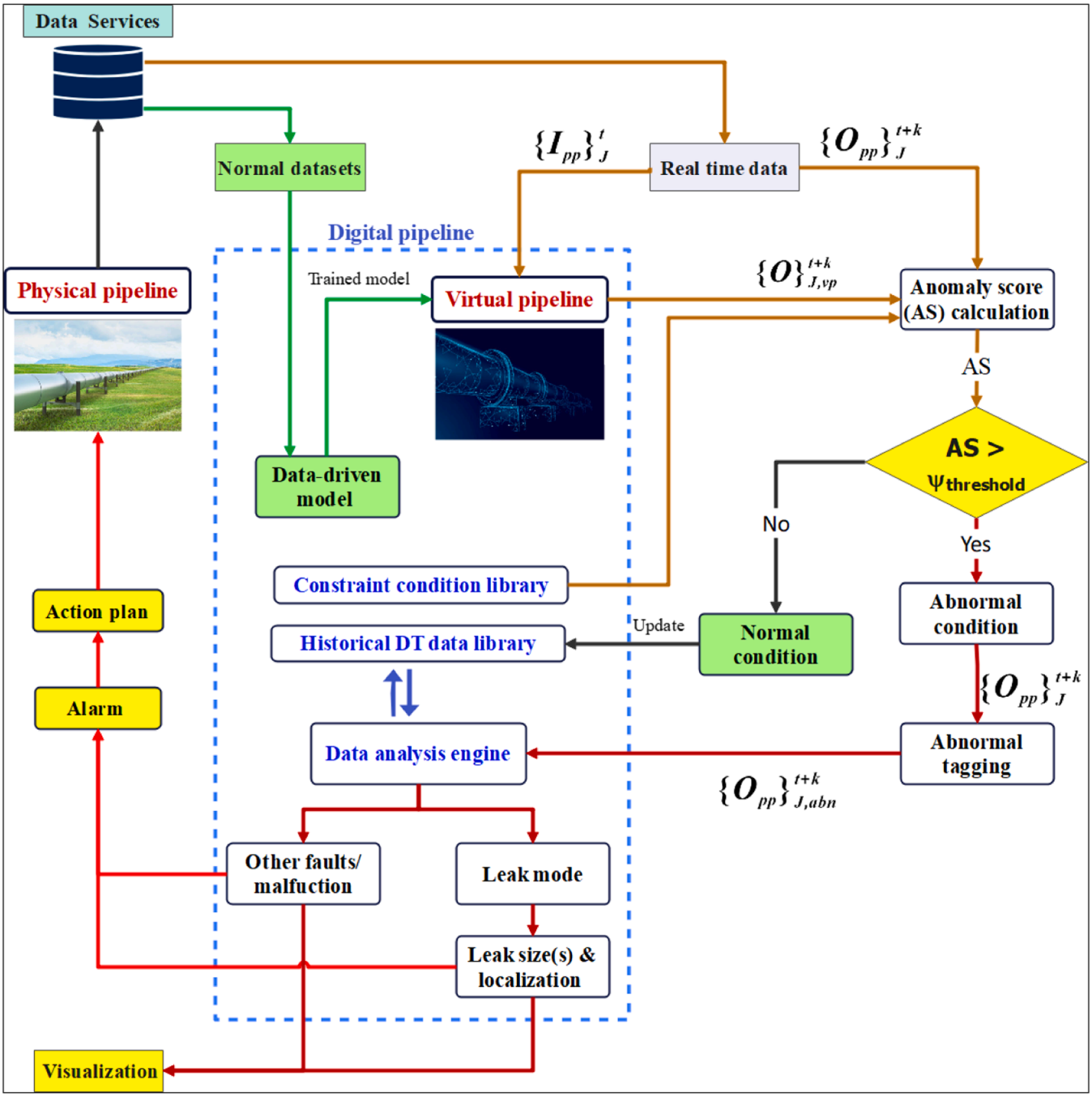


Fig. 8. Flow chart of the proposed pipeline DT model.

the VP ($\{O_{VP}\}_{J,i}^{t+k}$) with the real-time outputs from the PP ($\{O_{PP}\}_{J,i}^{t+k}$), the residuals between them ($R_{J,i}^{t+k}$) is defined as in Eq. 1. Then, the AS is defined as in Eq. 2.

$$R_{J,i}^{t+k} = \{O_{PP}\}_{J,i}^{t+k} - \{O_{VP}\}_{J,i}^{t+k} \quad (1)$$

$$AS_J = \sqrt{\frac{\sum_{i=1}^{n} R_{J,i}^{t+k}}{n}} \quad (2)$$

where k denotes the time lag of the pipeline, and n is the number of data points that are used in calculating the residuals. The AS should consider the various error sources that may indicate the pipeline status as leak while it is under normal operation. The primary sources of error in the proposed DT model for leak size and location estimation include sensor inaccuracies, data imbalance, and model assumptions. Variability in sensor calibration and noise in pressure and flow rate measurements can introduce deviations in estimation accuracy. Limited data for multiple-leak scenarios further impacts the model's generalization, while simplifications in physical and data-driven models, such as constant wave propagation speed or friction coefficients, may not fully capture the dynamic conditions of pipelines. Sensitivity analysis reveals that the model is robust to moderate variations in pipeline conditions, such as pressure levels and flow rates, but extreme changes, including rapid pressure drops or flow instabilities, increase the average relative error from 3.21 % to 7.5 %.

Table 7
Typical methods for denoising signals.

Denoising method	Features	Limitations
Moving averages [77]	Simple, effectively remove HF noise	Not suitable for applications where sudden, significant changes are of interest
Low-pass filters [78]	Attenuating HF noise	Complex, requires domain expertise
Wavelet transform (WT) [79]	Versatile, and can remove noise while preserving both high and low-frequency features in the signal	Computationally intensive
Empirical mode decomposition (EMD) [80]	Suitable for signals with non-linear and non-stationary components	Computationally intensive
Deep Learning Models (e.g. CNN, RNNs)	Can learn complex patterns in signals	Require large amounts of labeled data for training

4.3.5. Data analysis engine

If the AS is higher than the threshold value (), then the data are marked as abnormal operation. Thus, these data (real-time inlet and outlet parameters of the PP) are directed to the data analysis engine to detect the abnormality characteristics. For example, an ANN can be trained to detect single/multiple leaks in the gas pipeline as depicted in Fig. 11. Example for ANN-based model for leak detection and localization. As conducted by Kim et al. [45], data for leaks in gas pipelines can be simulated and generated at different locations and sizes using the OLGA platform (or other pipeline simulation platforms). These data are used to train the ANN model to be used later to detect leak size and location based on the abnormal data that are fed to the data analysis engine. However, the abnormal data may be obtained due to other abnormalities rather than leaks (such as component malfunctions). Thus, it is important to develop other models that can detect various types of abnormalities based on the degree of deviation between the VP outputs and PP outputs.

Different statistical metrics are typically used to evaluate the performance of the ANN model (or other statistical models) including the mean square error (MSE) and the coefficient of determination (R^2),

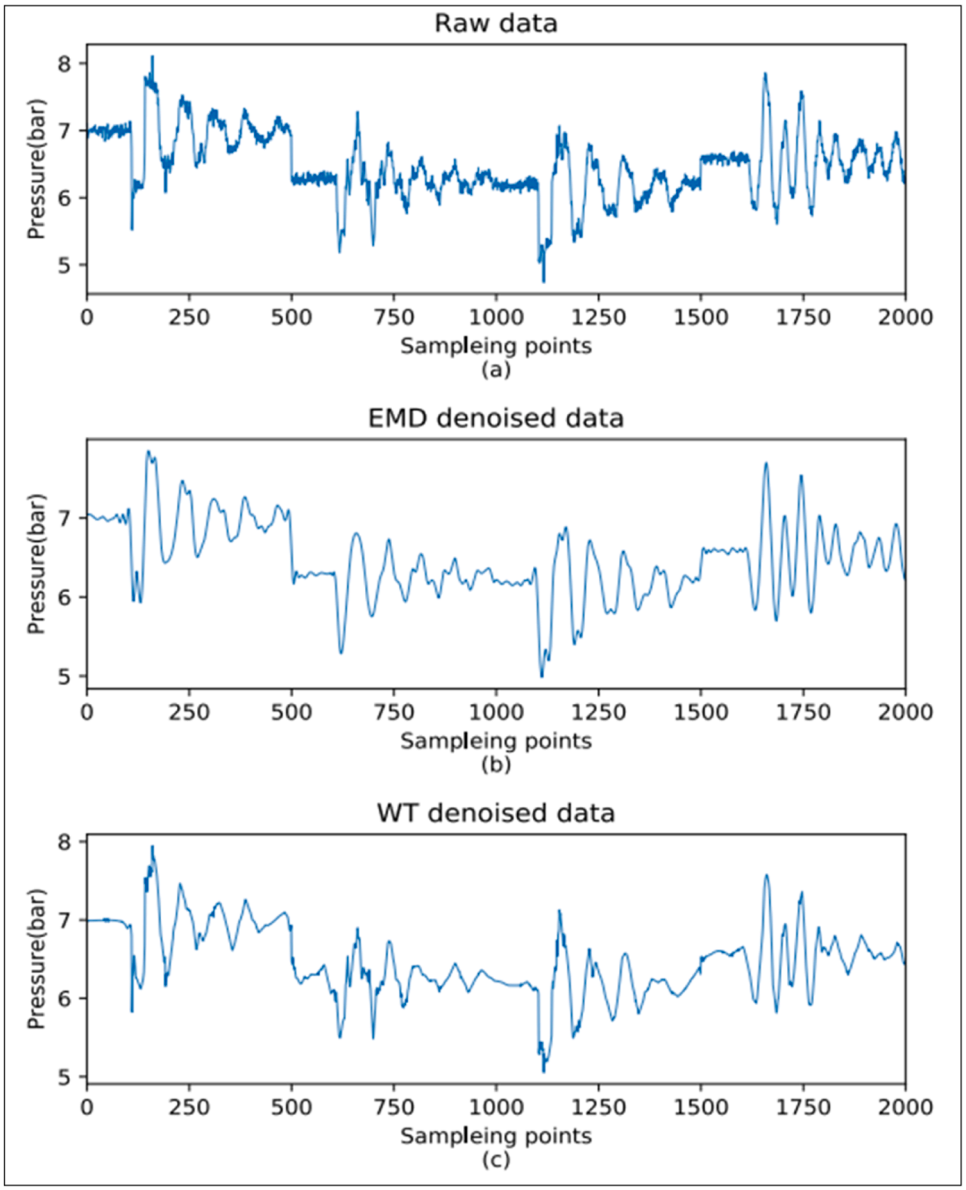


Fig. 9. Example of (a) raw pressure data and denoised data using (b) empirical mode decomposition, and (c) wavelet transform.

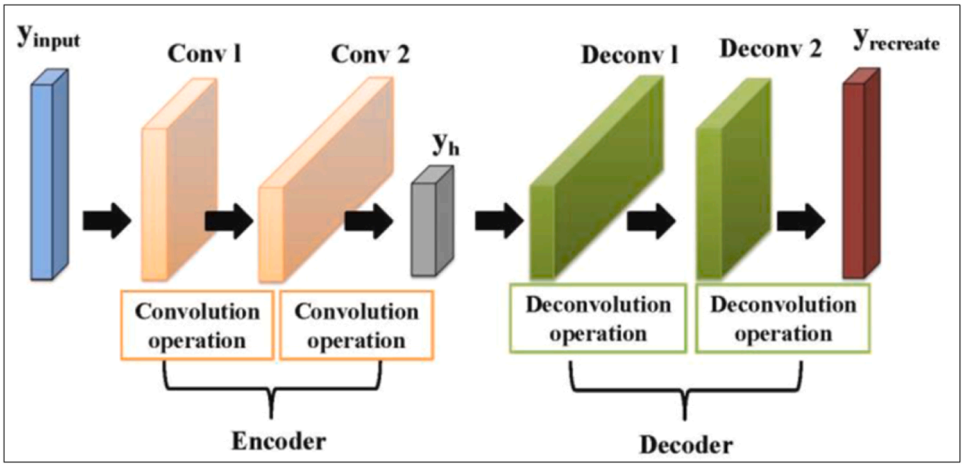


Fig. 10. Example for the virtual pipeline based on autoencoder-convolutional neural network (AE-CNN) model.

Table 8
Machine learning techniques can be used to develop the data-driven model of the virtual pipeline.

Technique	Features	Limitations
Linear Regression	Simple and interpretable.	May not capture complexities.
Decision Trees	Non-linear model capturing complex relationships.	Prone to overfitting, may not generalize well.
Random Forest	Ensembling decision trees reduces overfitting.	Complexity increases with the number of trees.
Gradient Boosting	Builds ensemble sequentially, robust, and accurate.	Sensitive to hyperparameter tuning.
Support Vector Machines	Effective in high-dimensional spaces.	Can be computationally expensive for large datasets.
Neural Networks	Captures complex relationships in data.	Requires large amounts of data and computational resources.
K-Nearest Neighbors	Simple and effective for small to medium-sized datasets.	Performance may degrade with high-dimensional data.
XGBoost	Scalable and accurate gradient boosting library.	Sensitive to hyperparameter tuning.
Autoencoder-CNN (AE-CNN)	Learns meaningful data representations through encoding and decoding.	May require substantial computational resources.

which are given as [81]:

$$MSE = \frac{1}{N}\sum_{I=1}^{N}(y_i - \hat{y}_i)^2 \tag{3}$$

$$R^2 = 1 - \frac{\frac{1}{N}\sum_{I=1}^{N}(y_i - \hat{y}_i)^2}{\frac{1}{N}\sum_{I=1}^{N}(y_i - \bar{y}_i)^2} \tag{4}$$

where y_i, $\hat{y}_i$, $\bar{y}_i$, and N are the target value (real value), the predicted value, the average value of the target values, and the number of data points, respectively.

4.3.6. Constrained condition library

Within the DT pipeline, the detection module monitors and identifies instances of undesired behavior, specifically patterns associated with leaks, through a comparison of the Virtual Pipeline (VP) output and Physical Pipeline (PP) output. Crucial to the leak determination process are the abnormal judgment threshold and diagnostic rules. The methods for calculating these thresholds encompass:

(i) Establishing a fixed threshold based on expert experience or the data distribution of reference samples. (ii) Designing a dynamic threshold using statistical methods.

(ii) The detection module selects the appropriate threshold calculation method based on the mission objective and response requirements, saving the computed threshold (Ψ) into the constraint condition library. The optimization of Ψ occurs based on the leak detection results (e.g., accuracy, fault detection rate (FDR), false alarm rate (FAR)) obtained from the benchmark dataset.

The assessment metrics for leak detection performance include accuracy, FDR, and FAR, which can be computed as follows [82]:

$$Accuracy = \frac{TP + TN}{Data_{all}} \tag{5}$$

$$FDR = \frac{TN}{Data_{leak}} \tag{6}$$

$$FAR = \frac{FN}{Data_{nor}} \tag{7}$$

where TP denotes the count of correctly identified normal data, $Data_{all}$ signifies the total number of all data, TN represents the accurately detected leak data, $Data_{leak}$ indicates the total number of leak data, FN signifies instances where normal data is incorrectly reported as leak data, and $Data_{nor}$ corresponds to the total number of normal data. In this scenario, the objective is to achieve high accuracy and FDR, while minimizing the FAR.

4.3.7. Malfunction diagnostic techniques

In the context of gas pipeline operations, the effective diagnosis of malfunctions is of paramount importance to ensure the integrity and reliability of the infrastructure. A spectrum of malfunctions, including leaks, corrosion, mechanical damage, and equipment failures, can potentially compromise the safety and operational efficiency of gas pipelines. To tackle these challenges, a suite of sophisticated diagnostic techniques has been developed, blending simulation tools and experimental methodologies. Pipeline modeling software (such as OLGA and Aspen HYSYS) facilitates the simulation of pressure changes and flow rates, while finite element analysis (FEA) tools like ANSYS and COMSOL help analyze structural stresses and potential weaknesses. Pipeline simulation tools such as Pipeline Studio offer a holistic approach to modeling the entire pipeline system, allowing for the introduction of simulated malfunctions, pressure fluctuations, and equipment failures.

In tandem with simulation tools, experimental tests play a pivotal

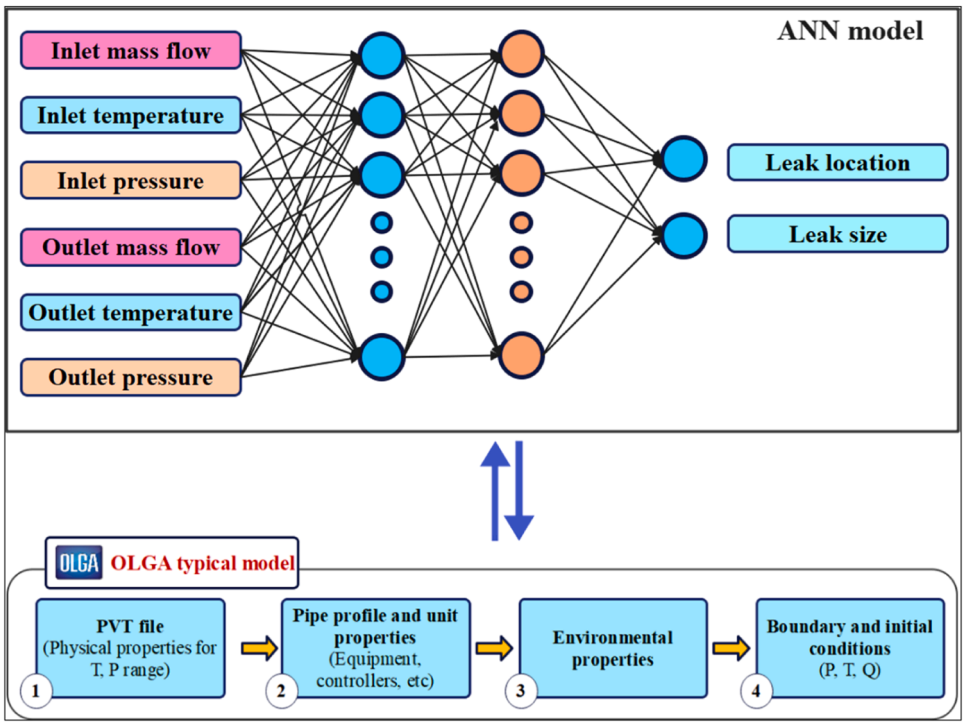


Fig. 11. Example of ANN-based model for leak detection and localization.

role in generating data that mirrors the characteristics of potential challenges. Controlled tests, including pressure testing, leak detection system testing, and cold flow testing, provide invaluable insights into the pipeline's response to varying conditions. These tests are complemented by the utilization of advanced technologies, such as drones equipped with sensors, to simulate visual inspections and detect anomalies. Moreover, emergency response drills involving various stakeholders serve as real-world exercises to simulate the actions and decisions required during a pipeline malfunction. Through these diagnostic techniques, operators can comprehensively understand the vulnerabilities of gas pipelines and implement proactive measures to enhance their robustness.

The generated data from these diagnostic techniques serves as the foundational building blocks for the development of DT models for gas pipelines. These DTs, virtual replicas of the physical pipeline systems, amalgamate data from simulations and experiments. Incorporating hydraulic parameters, structural stresses, dynamic simulations, and real-time sensor data, DT models facilitate a nuanced understanding of the pipeline's behavior under various scenarios. Anomaly detection algorithms, often integrated with machine learning, enable these DTs to identify deviations from normal operating conditions. The continuous refinement of these models through calibration and validation ensures their accuracy, empowering operators to monitor and predict abnormalities, optimize maintenance strategies, and bolster the overall resilience and safety of gas pipelines.

4.3.8. *Historical DT library*

The real-time tested data in the DT model that are classified as normal or abnormal conditions should be directed to the digital platform of the VP to update the DDM and the other abnormality detection models. Also, the data in this library can be scaled and used to develop trained DT models for other pipelines using transfer learning techniques.

5. Case study of leak detection abnormality

To evaluate the feasibility and effectiveness of the proposed DT model, a case study is conducted for leak detection and localization on a real-field offshore gas pipeline in this section. The detailed profile and flow characteristics of the pipeline are introduced in Section 5.1. Then, the data generation and validation of this case study are explained in Section 5.2. After that, the leak detection analysis is discussed in Section 5.3.

5.1. *Pipeline and flow characteristics*

Due to the limitation of abundant experimental data for leaks in gas pipelines, different leak scenarios can be generated using validated simulations for real-field pipelines. For the present case study, the pipeline profile (Fig. 12) of the offshore gas pipeline operated in the Bay of Bengal [45] is adapted here to investigate the feasibility of the proposed DT approach for leak detection. The pipeline length is 13 km with an internal diameter of 31.72 cm and the water depth is between 100 and 180 m. At normal operating conditions, this pipeline transports natural gas at a mass flow of 64.78 kg/s, inlet temperature of 52.9 °C, inlet pressure of 107.0 bar, and outlet pressure of 82.5 bar. The molar composition of the transported gas is 99.433 % methane, 0.239 % carbon dioxide, 0.163 % nitrogen, 0.084 % ethane, 0.018 % propane, 0.013 % butane, 0.005 % pentane, 0.002 % hexane, and 0.043 % of C7-heptane and above.

5.2. *Data generation and validation*

A data-driven model (DDM) is developed for the present pipeline by creating its profile in Aspen HYSYS with the aforementioned gas composition. Then, the pipe specifications are defined (internal diameter of 31.74 cm, roughness of 0.05 mm, ambient temperature of 12 °C, and overall heat transfer coefficient of 200 W/m^2-°C) and the thermodynamic properties of the gas are calculated using the Peng-Robinson equation of state. Further details of the pipeline parameters and simulation setting are provided in the supplementary material (sections S1 and S2). The gas flow is simulated over a wide range of inlet mass flow rates (from 10 to 80 kg/s), inlet pressure (70–120 bar), and inlet temperature (from 10 to 70 °C). The outlet pressure, temperature, and mass flow rate are collected for each inlet data set. A total number of 1161

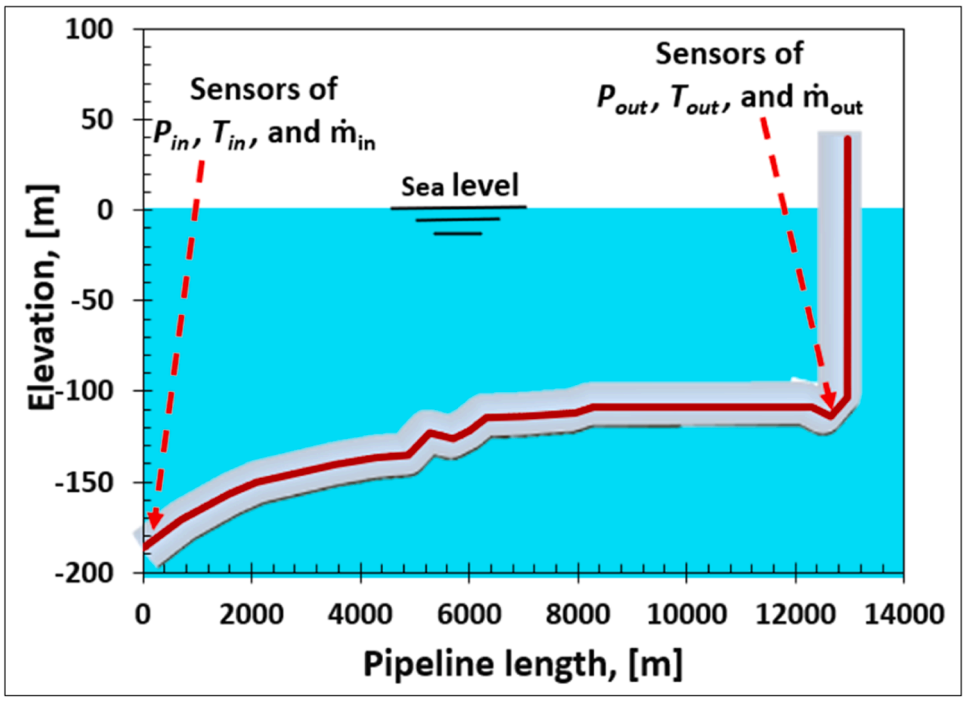


Fig. 12. Pipeline profile that is used in the present case study (adapted from [45]).

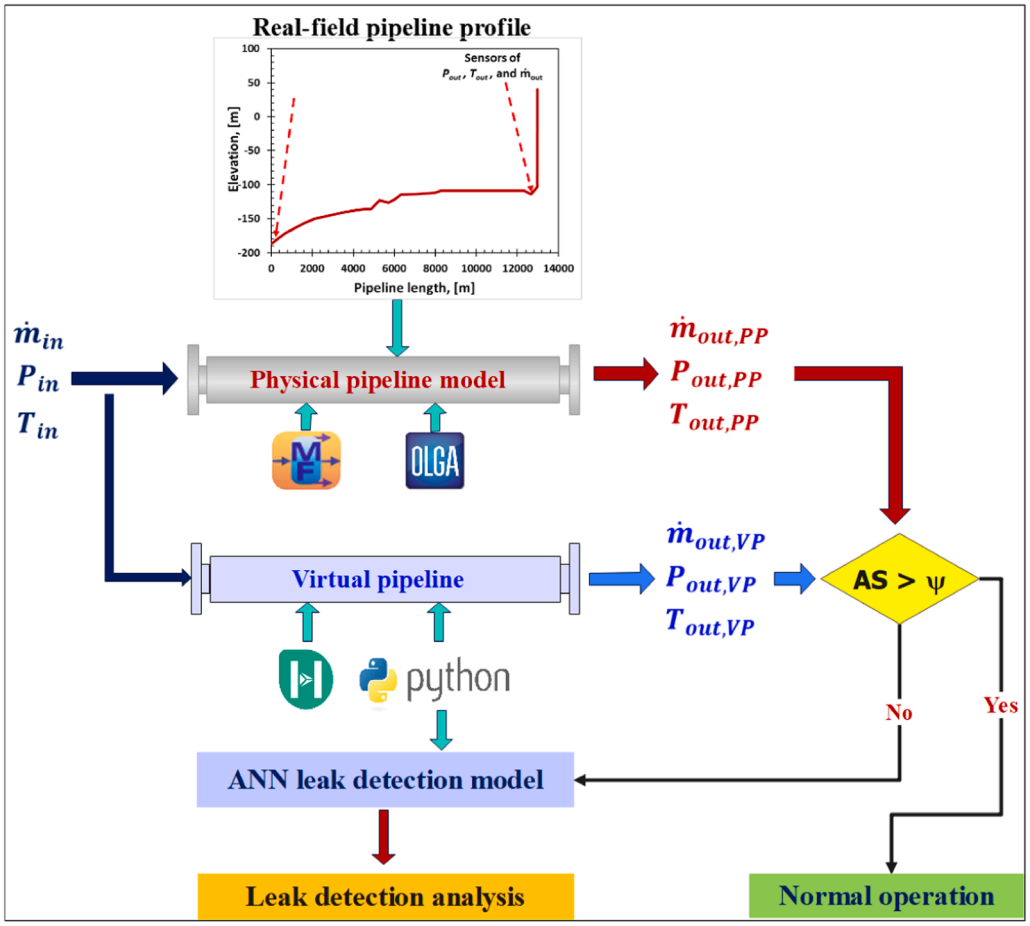


Fig. 13. Methodology of developing a DT model for leak detection for the gas pipeline profile of the present case study.

data sets are generated for the pipeline flow without any leak (healthy data or normal operating data ($Data_N$)). These data are used to train and test the data-driven model that forms the virtual pipeline in the DT model. The data-driven model is created in Python using the linear regression technique with 80 % of $Data_N$ used for model training and 20 % of $Data_N$ used for testing the performance of the model. As shown in Fig. 13, the data-driven model (virtual pipeline) will receive the inlet mass flow, pressure, and temperature from the inlet of the physical pipeline ($\dot{m}_{in}, P_{in}, T_{in}$) to predict the outlet parameters that should be obtained without any leak. These parameters are referred to as virtual pipeline outputs ($\dot{m}_{out,VP}, P_{out,VP}, T_{out,VP}$). The virtual pipeline outputs are compared to the physical pipeline outputs () by calculating the AS and comparing it to the threshold value (Ψ). If the AS < Ψ, then the data set is tagged as normal operation. If AS > Ψ, then this means that there is a leak (or multiple leaks) that needs to be detected in the leak-detection model (LDM).

For the LDM, forward ANN models are developed in Python to detect single or multiple leaks (with their size(s) and location(s) in the present pipeline. To train these models and due to the lack of actual leak-based data for this pipeline, several leak scenarios were created and simulated in OLGA software (see supplementary material Sections S1 to S3). For example, a single leak is created in the pipeline as x = 50 m, then the simulation is running to get the output parameters with this leak. Then, the position of the leak is moved to x = 100 m, and the simulation is performed again. This process is repeated until the leak position reaches x = 12950 m. All these simulations are also repeated at different leak sizes (from 0.5 cm to 5 cm with a step of 0.5 cm). Therefore, 2590 data sets were generated for a single leak at different positions and sizes.

It is important to mention that both DDM (in Aspen HYSYS) and LDM (in OLGA) are validated with normal operating data (from the real-field of the present pipeline profile as reported in [45], see Fig. 14(a)) and leak-based data from the experimental work conducted by Molina-Espinosa et al. [83] (Fig. 14(b)). As shown in Fig. 14(a), the simulated pressure profile matches the real-filed profile with a relative error of less than 1 %. The experimental data in Fig. 14(b) were obtained from measurements conducted on a water pipeline with a total length of 1.6 m, an inner diameter of 12.7 mm, and an inlet flow rate of 0.32 kg/s. The measurements were taken under two conditions: "no leak" (with a leak percentage, LP, of 0 %) and with a single leak positioned at x = 0.73 m from the inlet of the pipe. As depicted in Fig. 14(b), the size of the leak was varied to obtain data for two different LPs (17 %, and 31 %). In OLGA software, the pipeline profile was created as a straight pipeline, and the boundary conditions were set to match the real experimental conditions. Subsequently, the pressure profile data were generated in OLGA for both "no leak" and single leak conditions and plotted alongside the experimental data for comparative analysis. It is noteworthy that the OLGA data exhibit a commendable agreement with the experimental data, displaying a maximum absolute error of less than 3.04 %. This result underscores the capability of the OLGA model to generate leak-based data with sufficient accuracy for training ANN models.

5.3. Leak detection analysis

After training the DDM (using Random Forest Regressor (RFR) in Python), 232 normal data sets were used to test it. The results of the predicted parameters compared ($\dot{m}_{out,VP}$, $P_{out,VP}$, $T_{out,VP}$) to the actual parameters ($\dot{m}_{out,PP}$, $P_{out,PP}$, $T_{out,PP}$) are depicted in Fig. 15. The results show that the DDM accurately predict the output parameters with MSE of 0.019 and R^2 of 0.9947. However, further analysis for the DDM still needs to be trained with real-field healthy data subjected to noise, which is recommended for future work. The trained DDM is saved in the DT library to call it later to predict the virtual outputs in the DT model.

Similarly, after training the LDM (using ANN), 519 data sets (single leak at different locations and sizes) are used to test the trained LDM. The testing results are presented in Fig. 16. The R^2 of the ANN model is 0.99 with an average absolute relative error (ARE = |Predicted value – Actual value|/Actual value) of 3.21 %. The results show that the ANN model accurately predicts the leak location with a minimum difference of 0.10 m and a maximum difference of 700 m. Thus, further improvement is still needed for the model either by increasing the data for the training or investigating other machine learning techniques to predict the leak size and location. Therefore, future work will analyze the DT model with more accurate LDM and under several leak scenarios (single and multiple leaks). Herein, the trained ANN model archives acceptable ARE (3.21 %), which is enough to test the feasibility of performing the DT model to detect and localize leak abnormalities.

After training and testing the DDM and LDM, 76 data sets (38 normal and 37 abnormal) were randomly generated at “no leak” and “leak conditions” to test the performance of the proposed DT model. Then, these data are forwarded to the DT model. The performance of the DT model at different threshold values is presented in Fig. 17. It can be noted that the DT model successfully detects all abnormal points, thus the FDR is 1.0 for all threshold values. However, at low threshold values, some normal points were detected as abnormal points. Thus, different threshold values were examined until the FAR reached zero at a threshold value of 0.4. it is noted that the increase of the threshold values to higher than 0.4 will negatively affect the accuracy and the FAR of the model. This implies that the framework of the present DT model can be performed on real-field data to develop accurate DDM and LDM. Then, the optimal values of the threshold can be identified using a systematic trial-and-error testing approach as performed in this case

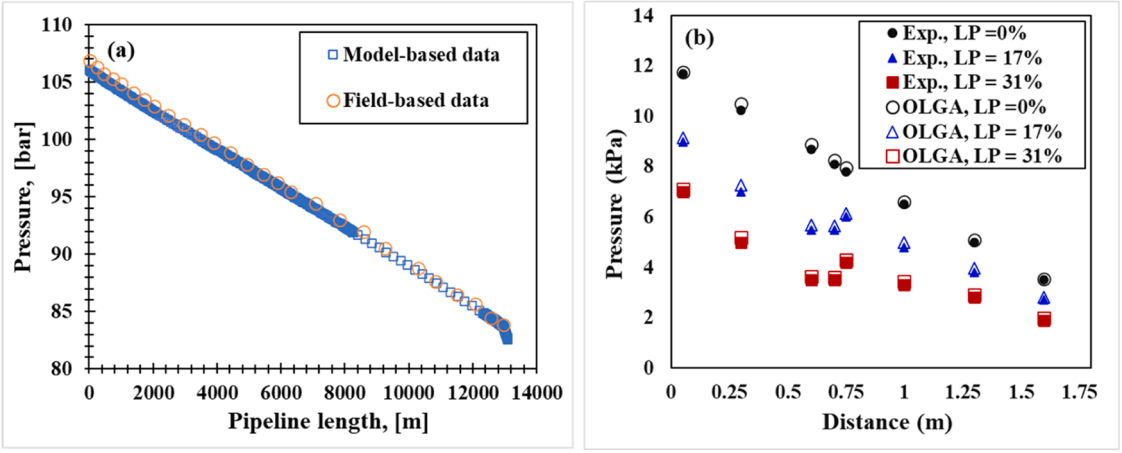


Fig. 14. Validation of the data generated in OLGA for (a) case study pipeline without leak, and (b) experimental pipeline with single leak. LP = Leak percentage.

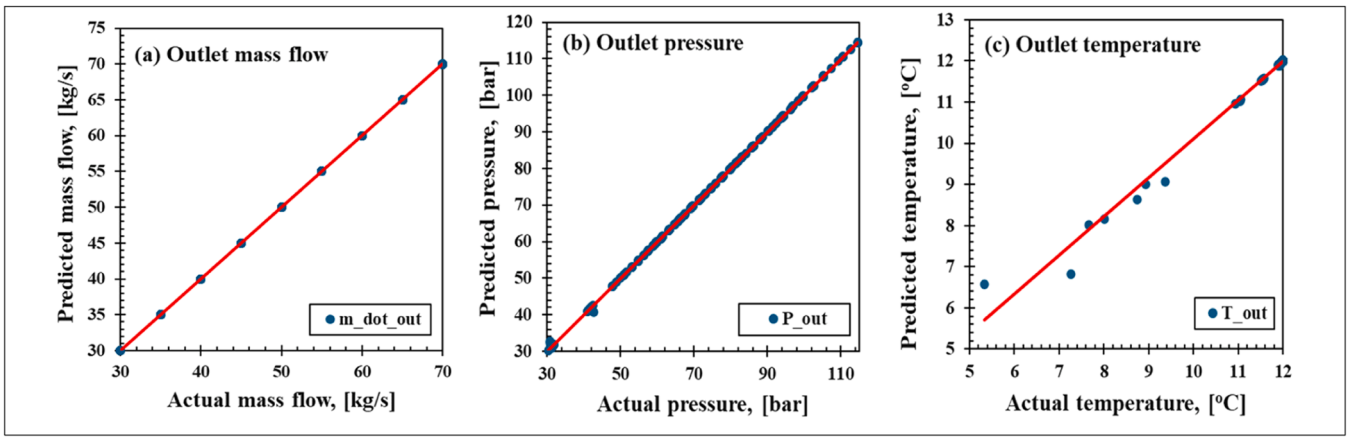


Fig. 15. Normal data set that was tested in the data-driven model to predict (a) outlet mass flow, (b) outlet pressure, and (c) outlet temperature (assuming there is no leak).

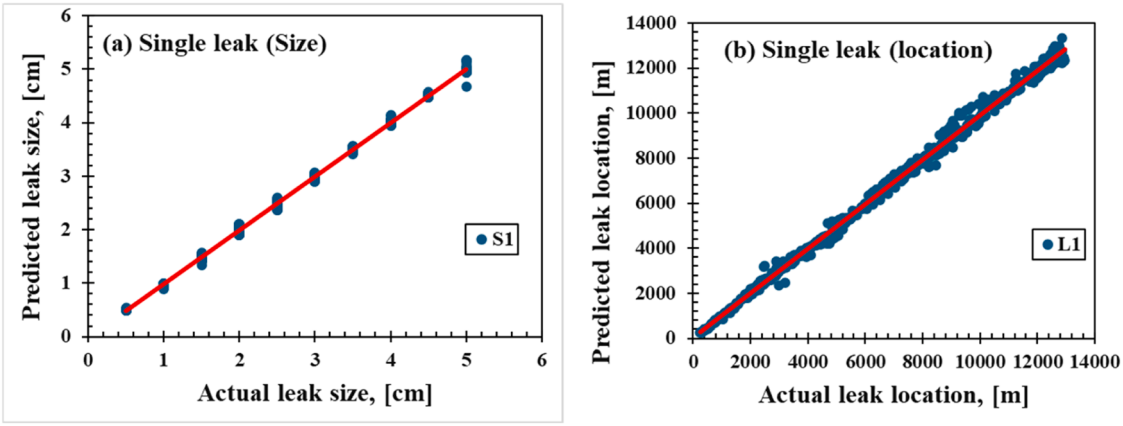


Fig. 16. Results of the abnormal data set of single leaks with different (a) sizes and (b) locations that were used to test the leak detection model. S1 = size of single leak, L1 = location of single leak.

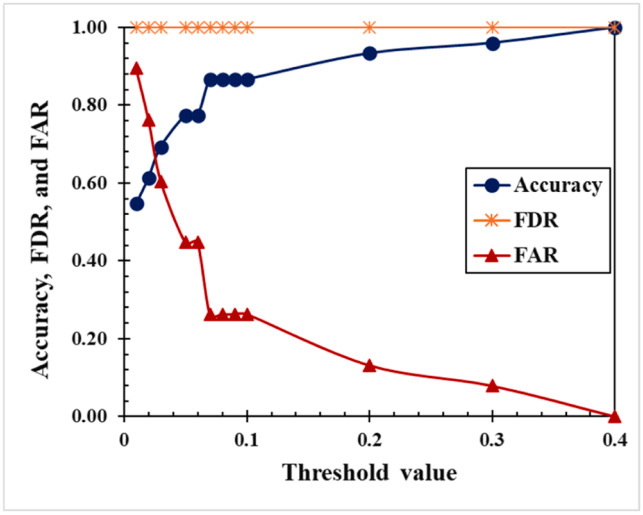


Fig. 17. Accuracy, FDR, and FAR at different threshold values.

Table 9
Comparison of performance metrics for the proposed DT model in this study with those available in the literature.

Study	Metric	Value	Notes
[70]	Detection Accuracy	91.23 %	Based on ELMD feature extraction with SVM and NB classifiers under low SNR conditions.
[72]	Detection Accuracy	> 85 %	Data-driven DT pipeline validated on simulated gas pipeline leak scenarios.
[67]	Prediction Accuracy	88 %	Predictive maintenance using LSTM for subsea pipeline DT system.
[68]	Error Rate	3.90 %	CVAE-based online real-time estimation of liquid pipeline leak parameters.
[69]	Localization Accuracy	Improved by 25 %	Fuzzy DT combined with backstepping algorithm and SVM for crack localization.
Proposed DT in this study	Detection Accuracy, ARE	100 % (no false alarms), 3.21 %	Validated under real-field and experimental scenarios with superior adaptability and scalability.

study.

Table 9 provides a detailed comparison of performance metrics from reviewed DT models and the proposed framework. Key metrics, including detection accuracy, error rates, and improvements in localization accuracy, showcase advancements across different approaches. Notably, the proposed DT achieves 100 % detection accuracy with zero false alarms and a low average relative error of 3.21 % for leak

detection and localization. The comparison underscores the versatility and superior performance of the proposed model, highlighting its ability to integrate seamlessly with real-field and experimental conditions while maintaining robust adaptability and precision. Also, the present DT model can be implemented for other pipelines as explained in the supplementary material (Section S5).

Despite the significant advancements of the proposed DT framework, certain challenges remain unaddressed, particularly concerning scalability and integration with IoT devices. Implementing the framework across extensive pipeline networks presents computational and data management challenges, necessitating efficient distributed computing architectures and advanced data aggregation techniques. Additionally, the integration of IoT sensor networks faces obstacles such as unreliable data transmission, limited energy efficiency, and reduced sensor reliability in harsh environments. Addressing these issues will require the development of energy-efficient communication protocols, robust sensor designs, and fault-tolerant algorithms. Future work will focus on incorporating edge computing for localized data processing, enhancing adaptive algorithms for scalable model deployment, and employing predictive maintenance strategies to improve sensor network reliability. These efforts will ensure that the DT framework remains robust, scalable, and efficient across diverse operational conditions and pipeline configurations.

6. Conclusions and recommendations

A comprehensive assessment of the hardware and software leak detection methods was presented in this study. It provides that most of these methods are either expensive or have poor performance with a high FAR. This emphasizes the need for a robust leak detection technique that can benefit from the abundant historical data of the existing pipelines to detect leaks with zero false alarms and inspect the future events of the pipeline leaks at a low cost. Therefore, several studies have been started to develop DT models that can present the behavior of the actual pipelines with high accuracy. From the present study, several conclusions are summarized as follows:

- Up to now, the trend of these models is to utilize machine learning techniques to form the virtual pipeline in the DT. Therefore, further studies are needed for DT models that utilize different modeling approaches for the virtual pipeline (data-driven model, physical-based model, or a combination of them).
- All of the presented DT models focus on pipeline leak detection without any information about their capabilities to identify the leak size and its location. Thus, these models need further improvement to provide full characterization of the leak events including leak size, and location under single or multiple leak scenarios.
- Some of the DT models were validated by experiments of a single leak in a short pipeline of compressed air. This type of validation did not reflect the multi-phase flow behavior in the oil/natural gas pipelines. Therefore, further experimental and theoretical analysis is still needed to develop DT models that capture single and multi-phase flow characteristics in these pipelines.
- The presented DT models are limited for leak detection. Therefore, the development of self-adaptive DT models that can detect any type of abnormalities during the pipeline operation (such as leak(s), equipment failure, mechanical damage, pressure fluctuations, etc.) is needed.
- The fast development of the trained pipeline DT model is challenging. Therefore, solutions for adapting the trained DT model from a particular pipeline to another pipeline are needed.
- The presented pipeline DT models were evaluated under steady-state pipeline operations. Therefore, a thorough evaluation of the pipeline DT model under transient conditions is needed. This analysis can be extended to cover several scenarios including (i) single and multiple leaks, (ii) single and multi-phase flow, and (ii) onshore and offshore pipelines.

Finally, from the presented case study in this review, the obtained results ensure that the pipeline DT model can be trained to detect leaks with zero FAR if the optimal value of the threshold is identified. However, this value is not constant and subjected to changes with the change of the conditions of the pipeline operation. Therefore, investigation of the pipeline DT model with dynamic threshold is also recommended for future work.

CRediT authorship contribution statement

Sleiti Ahmad K.: Conceptualization, Formal analysis, Funding acquisition, Investigation, Project administration, Supervision, Writing – review & editing. **Rahman Mohammad Azizur:** Funding acquisition, Project administration, Writing – original draft. **Rezaei-Gomari S.:** Writing – review & editing. **Hassan I.:** Writing – review & editing. **Hassan R.:** Writing – review & editing. **Al-Ammari Wahib:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. None

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.aej.2025.03.054.

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